Group A

1. Given the following Java program:

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        List num = new ArrayList(Arrays.asList(23, 16, 14, 33, 19, 6, 1));
        System.out.println("List is "+num);
    }
}
```

(a) Give the index values of all the odd numbers assuming zero-based indexing

ANSWER:

Odd numbers in array	Index
23	0
33	3
19	4
1	6

(b) How many elements would be looked at when the list is traversed (from start to finish) until the value 19 was found?

ANSWER:

When the list in the program is traversed, 5 elements in the array will be looked at.

2. Which of the following lists are syntactically correct in Java?

(a) List num = new ArrayList(Arrays.asList(1, 2, 3, 'four'));

ANSWER:

No.
Firstly, multiple characters or the string cannot be within single quotes. Strings are inside double quotes.

And, the ArrayList object needs to have similar datatype inside. If the above statement tried to compiled, it will have a compilation error telling that the ArrayList cannot be resolved to a datatype.

(b) List num = new ArrayList(Arrays.asList(1, 2, [3, 4]));

ANSWER:

The statement results in error. Because the syntax of defining array directly with the big brackets [] is not supported by java.

And even if there was correct syntax of array in that parameters, the asList method expects the arguments to be of similar data type which is not in this example. They are of type int and int[].

3. Perform a series of list operations on the following list:

List fruit = new ArrayList (Arrays.asList('apple', 'banana', 'pear', 'cherry'));

to produce this updated list:

['Grapefruit', 'banana', 'Date', 'cherry', 'Orange']

ANSWER:

```java
import java.util.*;
public class Test {
    Run | Debug
    public static void main(String[] args) {
        List<String> fruit = new ArrayList<String> (Arrays.asList(...a: "apple", "banana", "pear", "cherry"));
        fruit.set(index: 0, element: "GrapeFruit");
        fruit.set(index: 0, element: "Date");
        fruit.add(e: "Orange");
        System.out.println(fruit);
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL Run: Test

```
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>  & 'C:\Program Files\Java\jdk-19\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionM
 'C:\Users\Manoj Neupane\AppData\Roaming\Code\User\workspaceStorage\5922b7ff097dc94862f7c5a24ee880a8\redhat.java\jdt_ws\Introductory Programming and Problem Solvoing_aaa74b1d\bin' 'Test'
[Date, banana, pear, cherry, Orange]
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>
```

Group B

1. Find in array.

```java
import java.util.*;
public class App {
    Run | Debug
    public static void main(String[] args) {
        int num;
        System.out.println(x: "Enter the integer to check: ");

        Scanner scanner=new Scanner(System.in);
        num=scanner.nextInt();

        ArrayList <Integer> listInt=new ArrayList<>(Arrays.asList(...a: 1,2,3,4,5,7,10,23,32,43,545,6));
        boolean found=false;
        for(int i=0;i<listInt.size();i++){
            if(listInt.get(i)==num){
                System.out.println(x: "Present in the arrray.");
                found=true;
                break;
            }
        }
        if(!found) System.out.println(x: "Not present in array.");
    }
}
```

2. Average

```java
import java.util.*;

public class App {
    Run | Debug
    public static void main(String[] args) {

        ArrayList<Integer> physicsMarks = new ArrayList<>(Arrays.asList(
            ...a: 67,76,87,96,55,78,45,67,78,56,75,46,78,76,54,68,97,89,47));

        int total=0;
        for (int i = 0; i < physicsMarks.size(); i++) {
            total+=physicsMarks.get(i);
        }
        double average=total/physicsMarks.size();
        System.out.println(average);
    }
}
```

3. Reverse array.

```java
Run | Debug
public static void main(String[] args) {
    List<String> fruit = new ArrayList<String> (Arrays.asList(...a: "apple", "banana", "pear", "cherry"));
    List<String> reversed  = new ArrayList<>();
    for (int i = (fruit.size()-1) ;i>=0;i--) {
        reversed.add(fruit.get(i));
    }
    System.out.println("Original: "+fruit);
    System.out.println("Reversed: "+reversed);
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>  & 'C:\Program Files\Java\jdk-19\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMe
 'C:\Users\Manoj Neupane\AppData\Roaming\Code\User\workspaceStorage\5922b7ff097dc94862f7c5a24ee880a8\redhat.java\jdt_ws\Introductory Programming and Problem Solvoing_aaa74b1d\bin' 'Test'
Original: [apple, banana, pear, cherry]
Reversed: [cherry, pear, banana, apple]
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>
```

4. Find max in array.

```java
import java.util.*;
public class Test {
    Run | Debug
    public static void main(String[] args) {
        int[] nums = {-89,-78,4,5,23,-6,10};
        int max = nums[0];
        for (int num : nums) {
            if(num>max)max = num;
        }
        System.out.println("Max: "+max);
    }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvo
 'C:\Users\Manoj Neupane\AppData\Roaming\Code\User\workspaceStorage\5922b7ff097dc94
Max: 23
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvo
```

5. Check if array is sorted.

Test > src > App.java > ...

```java
import java.util.*;
public class App {
    Run | Debug
    public static void main(String[] args) {
        int []arr = {1,2,3,4,5,6,7,8,9} ;
        int []newarr=arr.clone();
        Arrays.sort(newarr);
        boolean sorted=false;
        if(Arrays.equals(arr, newarr)) sorted=true;
        if(sorted) System.out.println(x: "Sorted");
        else System.out.println(x: "Not sorted");
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
3d02250\redhat.java\jdt_ws\Test_14ec4a2e\bin' 'App'
Sorted
PS C:\Users\Fujitsu\OneDrive\Herald\Introductory Programing and Prob
```

Group C

1. Append to the end of hashset.

Since, HashSet class doesn't assure element synchronization, it cannot be guaranteed that element added goes to the last.

But hashtable is synchronized, so it can be done in hashtable.

Introductory Programming and Problem Solvoing > Test.java > Test > main(String[])

```java
Run | Debug
public static void main(String[] args) {
    Hashtable<Object,Object> table = new Hashtable<Object,Object>();

    table.put(key: "Name", value: "Manoj");
    table.put(key: "age", value: 19);
    table.put(key: "Description", value: "I am developer.");

    System.out.println(table);
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>  & 'C:\Program Files\Java\jdk-19\bin\ja
 'C:\Users\Manoj Neupane\AppData\Roaming\Code\User\workspaceStorage\5922b7ff097dc94862f7c5a24ee880a8\redhat.java\jdt_ws\Introdu
{Name=Manoj, age=19, Description=I am developer.}
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>
```

2. Java set intersection.

```java
public class Test {
    Run | Debug
    public static void main(String[] args) {
        Set<Integer> set1 = new HashSet<Integer>(Arrays.asList(...a: 1,2,3,4));
        Set<Integer> set2 = new HashSet<Integer>(Arrays.asList(...a: 3,4,5,6));
        Set<Integer> common = new HashSet<>();
        Iterator<Integer> set1iterator = set1.iterator();
        while(set1iterator.hasNext()){
            int num = set1iterator.next();
            if(set2.contains(num))common.add(num);
        }
        System.out.println("Common between two : "+common);
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>  & 'C:\Program Files\Java\jdk-19\bin\java.exe' '
 'C:\Users\Manoj Neupane\AppData\Roaming\Code\User\workspaceStorage\5922b7ff097dc94862f7c5a24ee880a8\redhat.java\jdt_ws\Introductory Pro
Common between two : [3, 4]
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>
```

3. Count pairs in hastable.

```java
import java.util.*;
public class Test {
    Run | Debug
    public static void main(String[] args) {
        Hashtable<String,Integer> marks = new Hashtable<>();
        marks.put(key: "Maths", value: 90);
        marks.put(key: "Programming", value: 85);
        marks.put(key: "Academic Skills", value: 64);

        System.out.println("The hastable contains "+ marks.size()+" key value pairs.");
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>  & 'C:\Program Files\Java\jdk-19\bin\java.exe' '--enable-preview' '
 'C:\Users\Manoj Neupane\AppData\Roaming\Code\User\workspaceStorage\5922b7ff097dc94862f7c5a24ee880a8\redhat.java\jdt_ws\Introductory Programming and Proble
The hastable contains 3 key value pairs.
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>
```

4. Collection view of map.

Introductory Programming and Problem Solvoing > Test.java > Test > main(String[])

```java
import java.util.*;
public class Test {
    Run | Debug
    public static void main(String[] args) {
        Map<String,Integer> marks = new HashMap<String,Integer>();
        marks.put(key: "Maths", value: 90);
        marks.put(key: "Programming", value: 85);
        marks.put(key: "Academic Skills", value: 64);

        System.out.println("\nThe collections view: "+ marks.values());
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>  & 'C:\Program Files\Java\jdk-19\bin\java
 'C:\Users\Manoj Neupane\AppData\Roaming\Code\User\workspaceStorage\5922b7ff097dc94862f7c5a24ee880a8\redhat.java\jdt_ws\Introduct

The collections view: [90, 64, 85]
PS C:\Users\Manoj Neupane\Desktop\Herald\Introductory Programming and Problem Solvoing>
```

Group D

1. Rock Paper Scissors.

Ask the user to enter in their move.

Make a list of valid moves.

Check if the user entered a valid move by looking at the list of valid moves. (If the move is in the list, it is valid move)

Randomly generate the opponent's move. (Randomly choose one move from the list of valid moves)

Display the result to user

Use a loop to continue asking the user for their move.

Check if the user wants to quit.

Introductory Programming and Problem Solvoing > Test.java > Test > main(String[])

```java
import java.util.*;
public class Test {
    Run | Debug
    public static void main(String[] args) {
        Random random = new Random();
        ArrayList<String> moves = new ArrayList<String>(Arrays.asList(...a: "R","P","S"));
        while(true){
            System.out.println(x: "\033[H\033[2J");//clears the screen
            String usermove = System.console().readLine(fmt: "Moves:\nR - Rock\nP-Paper\nS-Scissor\n\nEnter your move:").toUpperCase();
            if (!moves.contains(usermove)){
                System.out.println(x: "\nInvalid Move");
                continue;
            }
            String botmove = moves.get(random.nextInt(moves.size()));//randomly selecting the bot move
            System.out.println("Your move: "+usermove+"\nBot move: "+botmove);
            if (usermove.equals(botmove))  System.out.println(x: "\n\tTie!");// tie condition
            else if (isUserWinner(usermove, botmove))  System.out.println(x: "\n\tYou won!\n");
            else System.out.println(x: "\n\tYou Lost!");
            String choice = System.console().readLine(fmt: "Play Again? (Y/N): ");//asking if user wants to play again
            if(!choice.toUpperCase().equals(anObject: "Y")){//if user enters something other than Y/y the program ends
                System.out.println(x: "\n\nThank you for playing");
                break;
            }
        }
    }
    public static boolean isUserWinner(String usermove,String botmove){//returns if the user won the game.
        if(botmove.equals(anObject: "R"))return usermove.equals(anObject: "P");
        else if(botmove.equals(anObject: "P"))return usermove.equals(anObject: "S");
        else if(botmove.equals(anObject: "S"))return usermove.equals(anObject: "R");
        return false;
    }
```

Gameplay:

```
Moves:
R - Rock
P-Paper
S-Scissor

Enter your move:P
Your move: P
Bot move: R

        You won!

Play Again? (Y/N):
```

```
Moves:
R - Rock
P-Paper
S-Scissor

Enter your move:P
Your move: P
Bot move: R

        You won!

Play Again? (Y/N): y
```

```
Moves:
R - Rock
P-Paper
S-Scissor

Enter your move:S
Your move: S
Bot move: R

        You Lost!
Play Again? (Y/N):
```

```
Moves:
R - Rock
P-Paper
S-Scissor

Enter your move:S
Your move: S
Bot move: R

        You Lost!
Play Again? (Y/N): n


Thank you for playing
```

```
Moves:
R - Rock
P-Paper
S-Scissor

Enter your move:p
Your move: P
Bot move: P

        Tie!
Play Again? (Y/N):
```